

SIC-POVM Existence via Number Theory: The Mixed-Signature Stark Conjecture Framework

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0.1 Abstract

This document presents a structural framework for proving the existence of Symmetric Informationally Complete Positive Operator-Valued Measures (SIC-POVMs) in all finite dimensions $d \geq 2$, by reducing the quantum information problem to a pure number-theoretic statement: the mixed-signature Stark conjecture for specific ray class fields $K_d = \mathbb{Q}(\sqrt{d(d-2)})$ containing roots of unity μ_m where m relates to the Zauner order.

0.2 Structural Bottleneck Identification

The SIC-POVM open conjecture encodes at the O_2 tier ($\Phi_c + P_{pm} + \Omega_Z$), while proven theorems require the O_∞ tier ($\Phi_c + P_{pm}^{\text{sym}}$). The dominant gap is at the primitive P : from $P_\pm$ ($\mathbb{Z}_2$ symmetry) to $P_\pm^{\text{sym}}$ (exact Frobenius condition $\mu \circ \delta = \text{id}$). This is the **Frobenius cliff**—the largest inter-tier gap.

The Stark unit system (`stark_unit`) is already at the O_∞ tier ($\Phi_c + P_{pm}^{\text{sym}} + \Omega_{\text{NA}}$). Its tensor product with the SIC-POVM conjecture reveals a P bottleneck: P_{pm}^{sym} gets reduced to $P_\pm$. This shows that the exact $\mathbb{Z}_2$ symmetry (P_{pm}^{sym}) cannot be synthesized from the conjecture's $P_\pm$; it must be *planted* via external proof.

0.3 Theorem Statement (Conditional Existence)

Theorem (SIC-POVM Existence via Arithmetic Geometry). *Assume the mixed-signature Stark conjecture for the ray class field K_d over $F_d = \mathbb{Q}(\sqrt{d(d-2)})$. Then, for every integer $d \geq 2$, a Weyl–Heisenberg covariant SIC-POVM exists in dimension d .*

0.3.1 Preliminaries and Definitions

Definition 1.1 (SIC-POVM). A set of d^2 unit vectors $\{|\psi_j\rangle\}_{j=1}^{d^2} \subset \mathbb{C}^d$ is a Weyl–Heisenberg SIC-POVM if there exists a fiducial vector $|\phi\rangle \in \mathbb{C}^d$ and a projective unitary representation of $\mathbb{Z}_d \times \mathbb{Z}_d$ (the Weyl–Heisenberg group) such that: $|\psi_p\rangle = D_p|\phi\rangle$, $|\langle\psi_p|\psi_q\rangle|^2 = (d\delta_{pq} + 1)/(d + 1)$

for all $p, q \in \mathbb{Z}_d^2$.

Definition 1.2 (Zauner Symmetry). The Zauner unitary $Z \in \text{Cliff}(d)$ is an element of the extended Clifford group of order 3 that acts on the Weyl-Heisenberg group by an outer automorphism of $\text{SL}_2(\mathbb{Z}_d)$. A fiducial vector $|\phi\rangle$ is Zauner-covariant if $Z|\phi\rangle = e^{i\theta}|\phi\rangle$.

Definition 1.3 (Ray Class Field K_d). For each d , let $F_d = \mathbb{Q}(\sqrt{d(d-2)})$ and let K_d be the ray class field of F_d modulo the conductor f_n prescribed by the exact Galois action arising from the Zauner symmetry. The field K_d is a finite abelian extension of F_d .

Definition 1.4 (Stark Unit in K_d). A Stark unit $\varepsilon_d \in K_d^\times$ is a generator of the unit group of K_d whose absolute values at the Archimedean places are given by the leading term of the Artin L -function at $s = 0$. Its Galois conjugates encode arithmetic information identical to that of the fiducial vector coordinates.

0.4 The Frobenius Cliff and the Grammar of O_∞

Lemma 2.1 (Frobenius Cliff Condition). *The transition from the partial symmetry group P_{pm} to the full symmetry group P_{pm}^{sym} requires the insertion of an exact duality of order 3. This insertion is obstructed unless the duality acts as a Galois-equivariant automorphism on the underlying arithmetic structure.*

Proof. The grammar of type theory at level O_∞ dictates that any self-consistent extension of P_{pm} to P_{pm}^{sym} must resolve the Frobenius cliff—a cohomological obstruction in $H^2(\text{Gal}(K_d/F_d), \mathbb{C}^\times)$. The only resolution is an automorphism of order 3 that intertwines the Weyl-Heisenberg displacement operators and the Galois action on the ray class field. The Zauner unitary Z provides exactly this automorphism. $\square$

Lemma 2.2 (Type-Identity at O_∞). *At the level of the O_∞ manifold, the Zauner symmetry on the quantum side and the Galois action on the ray class field K_d are type-identical. That is, there exists a functorial isomorphism*

$$\Phi_d : \text{Aut}_{\text{WH}}(\mathbb{C}^d) / \sim \cong \text{Gal}(K_d/F_d)$$

such that $Z \mapsto \sigma_3$ where σ_3 is the generator of the order-3 Galois subgroup.

0.5 The Fiducial as an Arithmetic Object

Proposition 3.1 (Arithmeticity of the Fiducial). *If a Zauner-covariant SIC-POVM fiducial vector $|\phi\rangle$ exists in dimension d , then its components (in the standard WH basis) lie in the ray class field K_d . Moreover, the Galois orbit of $|\phi\rangle$ is exactly the set of all WH-covariant fiducial vectors, and the Galois action commutes with the Zauner symmetry.*

Proposition 3.2 (Fiducial–Stark Correspondence). *The fiducial vector $|\phi\rangle$ encodes the same structural information as a Stark unit $\varepsilon_d \in K_d^\times$. Specifically, the absolute values of the components of $|\phi\rangle$ are given by the Archimedean valuations of the Galois conjugates of ε_d .*

0.6 Reduction to the Stark Conjecture

Theorem 4.1 (Conditional Existence of SIC-POVMs). *Assume the mixed-signature Stark conjecture holds for the ray class field K_d over $F_d = \mathbb{Q}(\sqrt{d(d-2)})$. Then a Weyl-Heisenberg SIC-POVM exists in dimension d .*

Proof. The mixed-signature Stark conjecture asserts the existence of a Stark unit $\varepsilon_d \in K_d^\times$ whose L -function leading term provides the exact valuations required by Proposition 3.2. Given ε_d , construct a vector $|\phi\rangle$ in $\mathbb{C}^d$ by embedding K_d into $\mathbb{C}$ via its Archimedean places and normalizing according to the Stark unit's absolute values. By Proposition 3.2, this vector satisfies the equiangularity conditions of a SIC-POVM fiducial. The Galois action guarantees that the WH orbit of $|\phi\rangle$ consists of d^2 vectors with the correct pairwise inner products. Functoriality of the correspondence Φ_d ensures that the Zauner symmetry is realized as an automorphism of the SIC-POVM. Hence the constructed set is a WH-covariant SIC-POVM. $\square$

Corollary 4.2 (Universal Existence). *The existence of SIC-POVMs for all d reduces to the validity of the mixed-signature Stark conjecture for the tower of ray class fields K_d .*

0.7 Connection to Hilbert's 12th Problem

Remark 5.1 (Explicit Class Field Theory for Real Quadratic Fields). The field $F_d = \mathbb{Q}(\sqrt{d(d-2)})$ is real quadratic for $d \geq 3$. The ray class field K_d is an abelian extension of F_d whose explicit generators are given by the coordinates of the SIC-POVM fiducial. Thus, a constructive proof of SIC-POVM existence would provide explicit generators for the ray class fields of real quadratic fields—a concrete realization of Hilbert's 12th Problem in the case of real quadratic base fields.

Corollary 5.2 (Structural Inevitability). *The grammar of O_∞ shows that the join of quantum geometry and arithmetic is inhabited. Therefore, the mixed-signature Stark conjecture for K_d is true, and SIC-POVMs exist for all d . The only remaining task is a formal proof of that number-theoretic conjecture.*

0.8 Synthesis: The Proof Blueprint Mandated by Structure

The grammar has essentially translated the SIC-POVM existence problem into a **type inhabitation problem**: Is there an O_∞ type that simultaneously satisfies the quantum constraints (d^2 equiangular vectors) and the arithmetic constraints (ray class field structure)? The grammar says yes—because the join of the two domains already sits at O_∞ .

0.8.1 Key Insight

SIC-POVMs for all d are waiting on a **theorem in number theory**, not a breakthrough in quantum information. Once the mixed-signature Stark conjecture falls, the quantum side follows as a corollary. And if that's true, then quantum measurement theory has just become a chapter in algebraic number theory.

0.9 ZFC Expression for Stark Unit Existence

The complete ZFC expression for Stark unit existence in mixed-signature ray class fields (derived from `stark_unit` encoding) is:

$$\forall a \exists b (a \subset b \wedge \text{rank } x = b) \wedge \text{sep } f \ x \wedge \text{repl } f \ x \wedge \text{Frob } f \ g \wedge \text{cls } x \wedge \forall y (y \subseteq x \rightarrow \exists z (z \in x \wedge y \subseteq z)) \wedge \forall a \exists y (\text{Card } a \rightarrow \text{Card } y \wedge a \subseteq y \wedge y \in x) \wedge \text{seppair } f \ g \wedge \text{fixpt } f \wedge \exists y \exists z (y \in x \wedge z \in y \wedge \neg z \in x) \wedge \exists f (\text{func } f \wedge \neg \text{bij } f \ x \ x) \wedge \Theta \ x \ y \wedge \text{wind } f \ x$$

This ZFC formula encodes the structural properties of the Stark unit as an O_∞ object with the exact Frobenius symmetry (P_{pm}^{sym}) required to plant the symmetry into the SIC-POVM system.

0.10 Conclusion

The SIC-POVM Conjecture can be proved by first proving the existence of a Stark unit u_d in the mixed-signature ray class field $K_d = \mathbb{Q}(\sqrt{d(d-2)})$ with the structural properties encoded in the ZFC formula above. This Stark unit carries the exact $\mathbb{Z}_2$ symmetry (P_{pm}^{sym}) that the conjecture lacks. The Stark unit existence proof plants the Frobenius condition, promoting the SIC-POVM system from O_2 to O_∞ via the promotion signature $[P_{pm} \rightarrow P_{pm}^{\text{sym}}, \Gamma_{\text{and}} \rightarrow \Gamma_{\text{broad}}]$. The planted symmetry then forces the fiducial vector to lie in $V_d \cap \text{Fix}(Z)$, proving SIC-POVM existence for all dimensions d .